Unit CamS3

SKU:U174













Description

Unit CamS3 is a compact camera unit based on the **ESP32S3** module, supporting **WiFi cloud** image transmission. It is equipped with 8M PSRAM and 16M Flash memory, paired with a 2MP camera (OV2640), featuring a DFOV of 66.5°, capable of capturing photos at a maximum resolution of 1600x1200. The built-in microSD card slot can be used to store images and other information. The built-in PDM microphone allows for sound recording. The HY2.0-4P port at the bottom exposes the S3 internal D+/D- pins, supporting firmware updates or USB serial communication via the M5Grove2USB-C module. The onboard 2x1x3 pin headers expose the ESP32S3 program download port. The factory firmware provides complete functionality for shooting and uploading to EZData, allowing users to easily configure and implement remote free image monitoring. This product is suitable for applications in remote monitoring, time-lapse photography, industrial automation, and more.

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Unit CamS3 Getting Started Guide

This tutorial will introduce you to how to burn firmware to the M5CAMS3 device using Burner and its features.

Features

- Based on the ESP32-S3-WROOM-1-N16R8 module, supporting WiFi
- WiFi cloud EZDATA or serial transmission of image data
- Equipped with a 2MP camera (OV2640)
- Built-in microSD card slot
- Built-in PDM microphone, supporting sound recording
- Built-in programmable indicator light

Includes

- 1 x Unit CamS3
- 1 x Grove2USB-C
- 2 x Male-to-male breadboard wires
- 1 x HY2.0-4P Grove cable (20cm)
- 1 x LEGO adapter

Applications

- Remote monitoring
- Time-lapse photography
- Industrial automation

Specifications

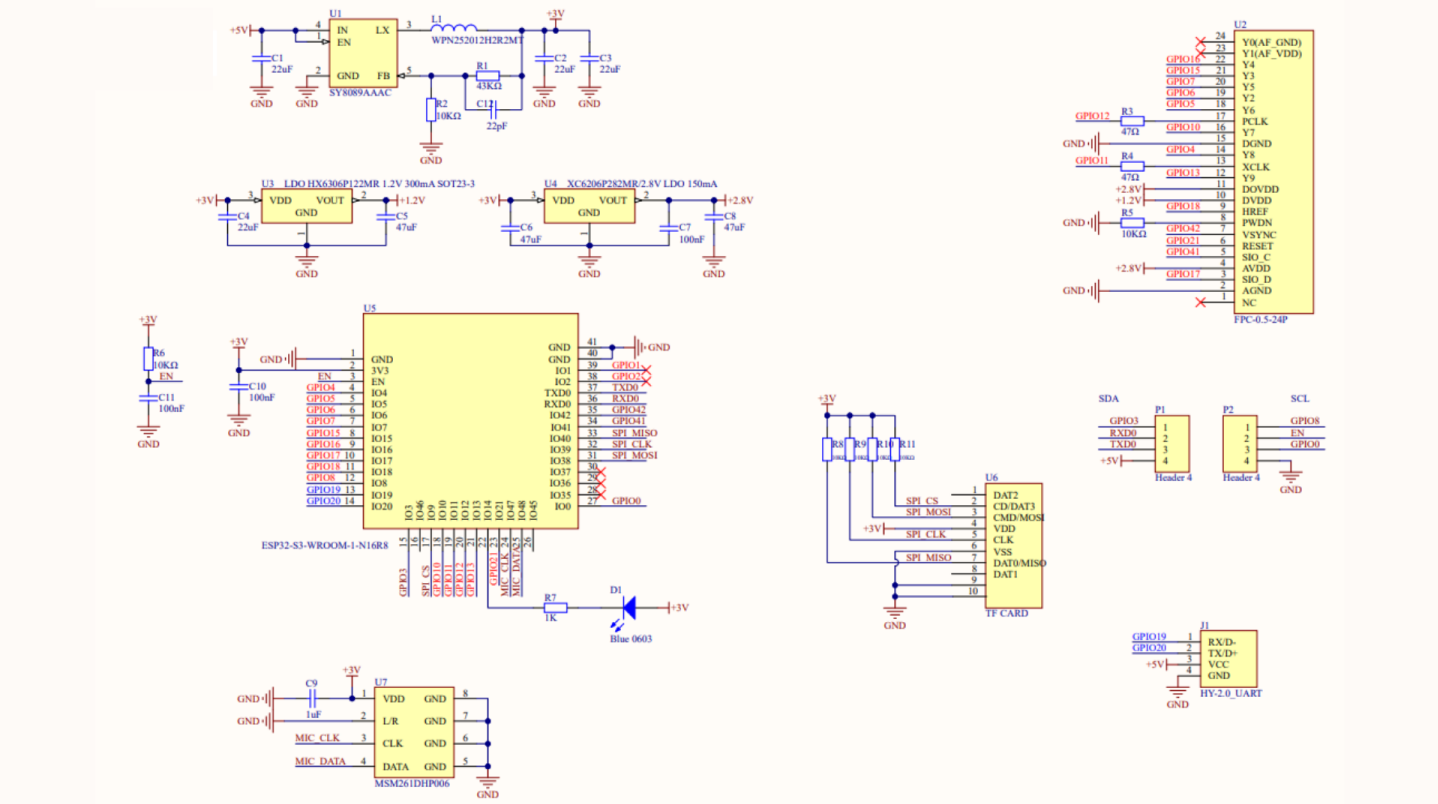
Specification	Parameter
ESP32-S3-WROOM-1-N16R8	Dual-core 240 MHz@ supports WiFi, onboard antenna
PSRAM	8M PSRAM
Flash	16M Flash
Camera	2MP@OV2640
OV2640 Output Formats	YUV (422/420) / YCbCr422, 8 bit compressed data, RGB565/555, 8/10 bit Raw RGB data
DFOV	66.5°
MIC	PDM microphone@MSM1261D4030HCPM
Memory Card	microSD card
Operating Temperature	0 ~ 40°C
Housing Material	Plastic
Product Size	40.0 x 24.0 x 11.0mm
Product Weight	8.3g
Package Size	138.0 x 93.0 x 12.0mm
Gross Weight	13.7g

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Download Program:

When using the Grove2USB-C provided in the product package for downloading, before powering on, short the G0 and GND pins of the pin header with the provided male-to-male breadboard wire to enter download mode, then use the Grove2USB-C for downloading.

Schematics



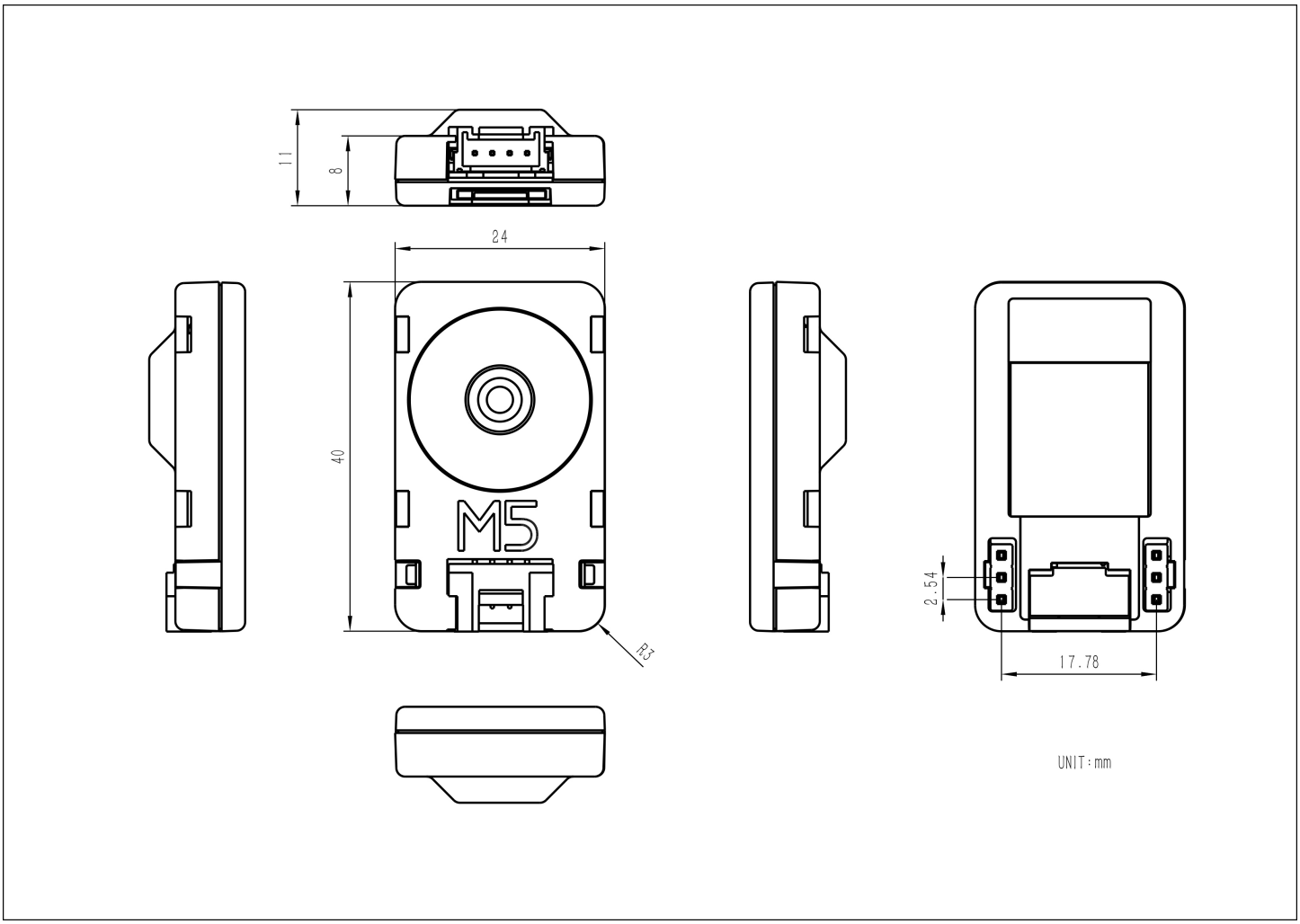
PinMap

Camera

Interface	Camera Pin	ESP32-S3
SCCB Clock	SIOC	G41
SCCB Data	SIOD	G17
System Clock	XCLK	G11
Vertical Sync	VSYNC	G42
Horizontal Reference	HREF	G18
Pixel Clock	PCLK	G12
Pixel Data Bit 0	D2	G6
Pixel Data Bit 1	D3	G15
Pixel Data Bit 2	D4	G16
Pixel Data Bit 3	D5	G7
Pixel Data Bit 4	D6	G5
Pixel Data Bit 5	D7	G10
Pixel Data Bit 6	D8	G4
Pixel Data Bit 7	D9	G13
Camera Reset	RESET	G21
Camera Power Down	PWDN	GND
Power Supply 3.3V	3V3	3V3
Ground	GND	GND

CAMS3 Unit	TX/D+	RX/D-	CS	MOSI	CLK	MISO	LED	MIC_CLK	MIC_DATA
GROVE	G20	G19							
SD Card Socket			G9	G38	G39	G40			
Programmable LED							G14		
MIC								G47	G48

Model Size



Datasheets

- OV2640
- ESP32-S3-WROOM-1-N16R8
- MSM1261D4030HCPM

Softwares

Arduino

- Unit CamS3 User Demo

EasyLoader

Easyloader	Download Link	Notes
Unit CamS3 Firmware Easyloader	download	/

After burning the firmware, connect your computer or phone to the "UnitCAMS3-WiFi" hotspot, enter 192.168.4.1 in the browser, and click "Stream" to view the image. Built-in EZData cloud for remote viewing, recording, SD card storage, and other practical functions.

Video

- Unit CAMS3 Remote Image Viewing and Time-lapse Photography

3c1447b3398128884d7bcbc0dce78fce.mp4

Product Comparison

	SoC	Memory	Camera	MIC	Download Mode
CAM Unit	ESP32-WROOM-32E	520 KB SRAM and 4 MB SPI flash	OV2640 2MP	/	ESP32-Downloader
CAMS3 Unit	ESP32-S3-WROOM-1-N16R8	8M PSRAM and 16M Flash	OV2640 2MP	MSM1261D4030 HCPM	ESP32-Downloader or Grove2USB-C

To compare information on the Unit CAM series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.